DATA MODELING

Reading: Chapter 7

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Data Modeling

- A technique for modeling data
- We've covered
  - RDM Model (e.g., Access Design View)
- The goal is to identify the structure of data to be stored in the database
- Data modeling is a partnership between the client and analyst

In industry, the job title is usually Business Analyst

The Building Blocks

- Entity
- Attribute
- Relationship
- Identifier

Entity name

Shares

code
name
price
quantity
dividend
pe

attributes

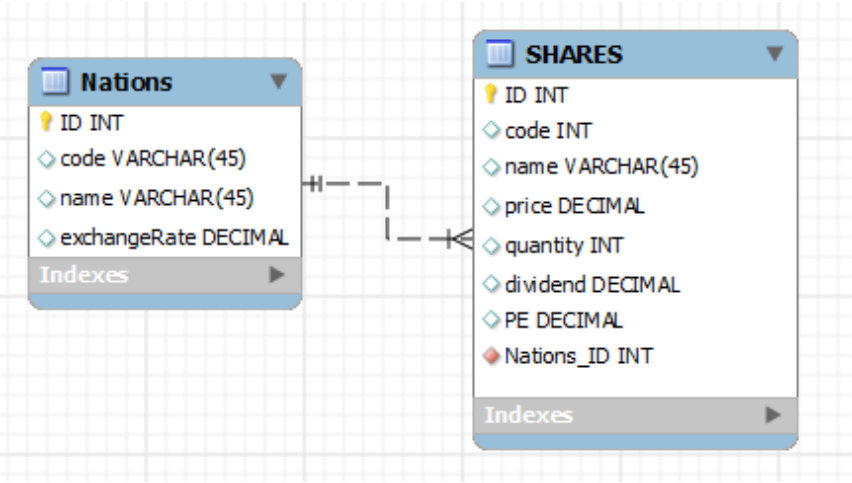
Remember our notation is slightly different from the text (e.g., plural entity name in upper camel case and singular attribute names in lower camel case)

A Well-Formed Data Model

- Follow organization (e.g., your company) convention
- No ambiguity
  - All entities, attributes, relationships, and identifiers are defined
  - Names are meaningful to the client

Relationships

- ERD and RDM show relationships between entities
  - 1-1
  - 1-many
  - Recursive
- ERD shows
  - Many-many
  - No foreign keys
- RDM usually shows
  - Associative entity (in-between table)
  - Foreign keys



Workbench uses more of a DB model style (not an ERD)

ERD Iteration

- You will usually iterate on your RDM model, as you learn the entities and relationships better
  - Is the level of detail correct?
  - Do the names represent the real-world entities and attributes?
  - Are the model components situated so that the design is clear (e.g., no hidden or crossing lines)
  - Are all exceptions handled?
  - Is the model accurate?

Are We on Track

Remember the “look for nouns” approach

- Write an RDM that provides a design for the following database statement:
  - The DB should store details about U.S. Presidents and their terms in office. Term in office can be limited to start year and end year. The DB should record details of their date of birth, gender and political party affiliation. Presidents are often associated with recent presidents in the era, so the DB should record the predecessor and successor of any president. The database should handle non-consecutive terms and should handle presidents whose political party changes.

Grover Cleveland served two-non-consecutive terms

Are We on Track

- Sketch a solution (in a group)
- Use a simple electronic tool (MS Word or Visio), and start with the RDM-Template.docx on the class Web site
- Be prepared to show your solution to the class (and iteratively refine the solution)

Are We on Track?

- An orchestra has four broad classes of instruments (strings, woodwinds, brass, and percussion). Each class contains musicians who play different instruments. For example, the strings section of a full symphonic orchestra contains 2 harps, 16-18 first violins, 14-16 second violins, 12 violas, 10 cellos, and 8 double basses. A city has asked you to develop a database to store details of the musicians in its three orchestras. All the musicians are specialists and play only one instrument for one orchestra.

Are We on Track?

- A travel agency is frequently asked questions about tourist destinations. For example, customers want to know details of the climate for a particular month, the population of the city, and other geographic facts. Sometimes they request the flying time and distance between two cities.